# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: Cryptography and Network Security**
**COURSE CODE: CSA51**

**COURSE OUTCOME COVERED**

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

**QUESTIONS: 5** **Total Marks:50**

Annexure A

## APPLICATION- BASED QUESTIONS

***Code of Conduct:***

*I* TALAPALA PARTHA SARATHI *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** ***TALAPALA PARTHA SARATHI***

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Annexure A

# APPLICATION- BASED QUESTIONS

**1. Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

**2. Scenario: Intercepted Suspicious Message**
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

**3. Scenario: Secure Messaging System**
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

**4. Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

**5. Scenario: Legacy Encryption in Corporate Communication**
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

Annexure A

# ANSWERS TO APPLICATION- BASED QUESTIONS

## Answer 1. Event Communication Security (Caesar Cipher)

The method is the **Caesar (shift) cipher**, a monoalphabetic substitution in which every letter moves a fixed number of places along the alphabet. With A = 0 ... Z = 25 and a key of k = 4:

$$C = (P + 4) \bmod 26$$

### a. Encrypted message for a shift of 4

| Plaintext | M | E | E | T | A | T | N | O | O | N |
|---|---|---|---|---|---|---|---|---|---|---|
| P (0-25) | 12 | 4 | 4 | 19 | 0 | 19 | 13 | 14 | 14 | 13 |
| P + 4 | 16 | 8 | 8 | 23 | 4 | 23 | 17 | 18 | 18 | 17 |
| Ciphertext | Q | I | I | X | E | X | R | S | S | R |

**Keeping the word spacing, MEET AT NOON is encoded as: QIIX EX RSSR**

### b. Retrieving the original message

An authorized member who knows the rule (Caesar cipher, shift 4) simply reverses it by moving every letter 4 places **back**: P = (C − 4) mod 26. Letters near the start of the alphabet wrap around (for example, E − 4 = 4 − 4 = 0 = A).

| Ciphertext | Q | I | I | X | E | X | R | S | S | R |
|---|---|---|---|---|---|---|---|---|---|---|
| C (0-25) | 16 | 8 | 8 | 23 | 4 | 23 | 17 | 18 | 18 | 17 |
| C − 4 | 12 | 4 | 4 | 19 | 0 | 19 | 13 | 14 | 14 | 13 |
| Plaintext | M | E | E | T | A | T | N | O | O | N |

**Recovered message: MEET AT NOON**

## Answer 2. Intercepted Suspicious Message (Atbash Cipher)

### a. Most likely original message

The text is a monoalphabetic substitution that mirrors the alphabet (A↔Z, B↔Y, C↔X ...), known as the **Atbash cipher**. It needs no key: P = 25 − C.

| Cipher letter | G | S | R | H | Z | V | X | I |
|---|---|---|---|---|---|---|---|---|
| Position C | 6 | 18 | 17 | 7 | 25 | 21 | 23 | 8 |
| 25 − C | 19 | 7 | 8 | 18 | 0 | 4 | 2 | 17 |
| Plain letter | T | H | I | S | A | E | C | R |

| Ciphertext | GSRH | RH | Z | HVXIVG |
|---|---|---|---|---|
| Plaintext | THIS | IS | A | SECRET |

**Original message: THIS IS A SECRET**

### b. Justification

- **Word shapes:** the pattern is 4 letters, 2 letters, 1 letter, 6 letters. The only one-letter English words are A and I, so Z must be one of them.
- **Repeating pattern:** the 2-letter word RH occurs twice and the 4-letter word GSRH ends with the same two letters. The shape xxRH RH Z fits THIS IS A, a very common English opening, so RH = IS.
- **Letter structure:** in HVXIVG the letter V repeats at positions 2 and 5, matching SECRET where E repeats (V = E).
- **One consistent rule:** every pair satisfies plain + cipher = 25 (Z→A, V→E, X→C, I→R, H→S, G→T, S→H, R→I). A Caesar shift is ruled out because Z→A is +1 while G→T is +13.
- **Cipher characteristics:** Atbash keeps word lengths and letter frequencies, is self-reciprocal (applying it twice returns the text), and produces meaningful English, which confirms the result.

## Answer 3. Secure Messaging System (Vigenère Cipher)

### a. Step-by-step encryption of HELLO with the keyword KEY

**Step 1:** Repeat the keyword under the message until it is as long as the message (KEYKE). **Step 2:** Convert letters to numbers (A = 0 ... Z = 25). **Step 3:** Add the key number to each plaintext number modulo 26.

$$C = (P + K) \bmod 26$$

| Position | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Plaintext** | H | E | L | L | O |
| **Repeated key** | K | E | Y | K | E |
| **P (number)** | 7 | 4 | 11 | 11 | 14 |
| **K (number)** | 10 | 4 | 24 | 10 | 4 |
| **P + K** | 17 | 8 | 35 | 21 | 18 |
| **(P + K) mod 26** | 17 | 8 | 9 | 21 | 18 |
| **Ciphertext** | R | I | J | V | S |

Only the 3rd letter wraps around the alphabet (35 − 26 = 9 = J). Encrypted word: **RIJVS**. Check: (C − K) mod 26 gives R−10 = H, I−4 = E, J−24 = L, V−10 = L, S−4 = O, i.e. HELLO.

### b. Influence of the repeating keyword

- **Each key letter is a different Caesar shift:** K, E and Y act as shifts of 10, 4 and 24. Character 1 uses K, character 2 uses E, character 3 uses Y, and then the cycle restarts (K for character 4, E for character 5).
- **Same plaintext letter, different ciphertext:** the two L's of HELLO become J and V because they meet different key letters (Y and K). Letter-frequency patterns are therefore hidden, which is why it is called a polyalphabetic cipher.
- **Key length sets the period:** with a 3-letter key the shifts repeat every 3 characters, so equal plaintext letters that are a multiple of 3 positions apart encrypt identically. Kasiski examination and the index of coincidence exploit this repetition to find the key length.

## Answer 4. Suspicious Digital Asset Investigation (Steganography)

### a. Detecting hidden data in the logo image

Hidden data in an ordinary-looking image indicates **steganography**, typically least-significant-bit (LSB) embedding or data appended to the file. The analysis is done on a bit-for-bit forensic copy whose SHA-256 hash is recorded first to preserve the chain of custody.

| Method | What the investigator does | Indicator of hidden data |
|---|---|---|
| **Baseline comparison** | Compare with a known clean copy of the logo (hash, file size, pixel difference) | Different hash or size, scattered pixel differences, no visible change |
| **File structure and metadata** | Hex editor, exiftool, strings, binwalk; check end markers (PNG IEND, JPEG FFD9) | Bytes after the end marker, unknown chunks, odd EXIF or comments, embedded files |
| **Bit-plane inspection** | Extract LSB planes of each colour channel (StegSolve, zsteg) | Structured or text-like pattern where random noise is expected |
| **Statistical steganalysis** | Chi-square, RS and sample-pair analysis; histogram plot | Equalized frequencies of value pairs (2k, 2k+1); high estimated embedding rate |
| **Automated tools** | StegExpose, Stegdetect; try steghide or outguess extraction | Image flagged, or data extracted using a weak or default passphrase |

Several independent signals together give a reliable conclusion; a single weak indicator only justifies deeper analysis.

### b. Two techniques to strengthen hidden communication

- **1. Encrypt (and compress) the payload before embedding.** Using a strong cipher such as AES with a high-entropy key makes the hidden bits look like random noise. Even if the steganography is detected, the extractor obtains only ciphertext, so confidentiality does not depend on the image staying undetected.
- **2. Use keyed, adaptive embedding at a low rate.** A secret key seeds a pseudo-random choice of pixels, preferably in textured or edge regions (content-adaptive schemes such as HUGO, WOW or S-UNIWARD), with matrix embedding to change fewer pixels. This removes the sequential pattern that chi-square and RS analysis rely on, and without the key the embedding positions cannot be found.

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

**Answer 5. Legacy Encryption in Corporate Communication (Rail Fence Cipher)**

**a. Original message**

Three levels of arrangement = **rail fence cipher with 3 rails**. The ciphertext has 25 letters; the zigzag period is 2(3−1) = 4, so rail 1 holds 7 letters (positions 0, 4, 8 ...), rail 2 holds 12 and rail 3 holds 6. Splitting the ciphertext: **Rail 1** = WECRLTE, **Rail 2** = ERDSOEEFEAOC, **Rail 3** = AIVDEN. Placing them back on the zigzag path:

| **Pos.** | **0** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Rail 1** | W | | | | E | | | | C | | | | R | | | | L | | | | T | | | | E |
| **Rail 2** | | E | | R | | D | | S | | O | | E | | E | | F | | E | | A | | O | | C | |
| **Rail 3** | | | A | | | | I | | | | V | | | | D | | | | E | | | | N | | |

**Reading the columns from left to right: WEAREDISCOVEREDFLEEATONCE → WE ARE DISCOVERED FLEE AT ONCE**

**b. Why the method is vulnerable today**

- **Letters are not changed**, only moved, so normal English letter frequencies stay visible and immediately reveal a transposition cipher.
- **Tiny key space:** the key is just the number of rails (and an offset). For a 25-letter message fewer than 24 values are possible, which a computer brute-forces in milliseconds, and each attempt is checked automatically against dictionary words.
- **Known-plaintext weakness and no modern properties:** predictable words or headers expose the key, and the scheme gives no diffusion, confusion, integrity or authentication.

**c. Practical enhancement**

Replace the transposition with a modern authenticated cipher such as **AES-256 in GCM mode**, with the key exchanged over TLS 1.3. A 256-bit key removes the guessable key, and GCM adds integrity and authentication so tampering is detected. As an interim step, combine the rail fence with a substitution (e.g. Vigenère with a long key) before migrating.

## RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |

| | | | | | |
|---|---|---|---|---|---|
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation | Basic analysis with limited insights | Poor or incorrect analysis of results |
| Documentation | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation |